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Optimal Retail Return Policies with Wardrobing

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Many retailers offer refunds to consumers who, after a trial period, return a product that they find does not fit their needs. Some consumers are willing to use this return option opportunistically for short-term consumption rather than its intended purpose of resolving fit uncertainty. Such behavior has been termed "wardrobing." Restocking fees (partial refunds) can be used to combat wardrobing. However, there is a trade-off involved, since partial refunds will be viewed negatively by consumers who return an item due to a true lack of fit. In this study, we consider how the extent of wardrobing (how many consumers consider such behavior) and the benefit of wardrobing (how much value can be extracted during the trial) impact firm pricing decisions and profits in this retail context. Our results imply that an increase in the extent of wardrobing is most detrimental to profits when the current extent of wardrobing is low. On the contrary, if the extent of wardrobing is already very high, and the benefit of wardrobing to consumers is also high, the retailer can set prices and refunds such that additional wardrobing actually increases firm profits. In a model extension, we show how a retailer can effectively screen wardrobers from ordinary consumers by offering a menu of price/refund pairs, and that such an approach can lead to increased profits if the extent of wardrobing is sufficiently high. Overall, our findings provide new insights into how retailers can set prices and refund policies to effectively manage opportunistic behavior by consumers.

Key words: retail operations; consumer returns; opportunistic behavior; wardrobing
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1. Introduction

Consumer returns represent a substantial challenge and a sizeable expense for retailers. In 2015, U.S. retailers are estimated to have accepted returns worth $260.5 billion, an increase of roughly 50% from 2007 (National Retail Federation 2008, Ng and Stevens 2015). The return rate is also on the rise, from below 8% to above 12%, in the past decade (National Retail Federation 2008, 2016), and the consulting firm Accenture estimates that return processing costs equate to about 4% of a typical retailer's sales (Douthit et al. 2011). Despite the substantial costs and complexities, providing consumers with an opportunity to return a non-defective retail purchase for a refund (possibly net of a restocking fee) is viewed as necessary by many retailers (Brill 2015). In general, the rationale for returns is the idea that it is often impossible for a consumer to fully resolve all uncertainty regarding the fit of the item to her needs before a purchase is made (Nelson 1970). In-store assistance, online reviews, and other resources can provide some information, but a trial period is often needed to assess fit. Indeed, Ferguson et al. (2006) explain that products are often returned with no functional or cosmetic defect, but rather because a mismatch between product attributes and consumer preferences is revealed during the trial period. Reasons for mismatch can include installation difficulty, product performance issues, or remorse (Kumar et al. 2002).

Our study is motivated by the observation that consumer returns are not driven exclusively by mismatch. Instead, some consumers take advantage of lenient return policies by *purchasing items with the explicit intent of returning them* after a period of use. For these consumers, the return option is essentially used as a rental opportunity. Such "renting via the sales channel" is well-known among retailers and is commonly referred to as "opportunism" or "wardrobing," the latter term derived from the frequent use of

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such tactics to purchase and return clothing items. In fact, men's clothing for special events was cited by Timberlake (2013) as a "prime target," along with jewelry, tools, seasonal items, and video cameras, for wardrobing behavior by opportunistic consumers (see Table 3 of Chu et al. (1998) for more examples).[1] A recent industry survey focusing on wardrobing in the women's clothing category shows that *one sixth* of the surveyed population admit to having engaged in it in the past (Osborne 2013). From the retailers' perspective, National Retail Federation (2014) reports that 72.7% experienced wardrobing in 2014, an increase of over 17% from the prior year, which is a significant contributor to the estimated $10.9 billion in annual retailer losses due to return fraud.

Wardrobing becomes possible when several common elements of the retail context align. Specifically, wardrobing requires a *trial period* combined with a return opportunity *in the sales channel*. These two common retail practices are intended to give consumers the opportunity to resolve the uncertainty in their idiosyncratic valuations of the product, but they also enable wardrobing by opportunistic consumers. Moreover, for many products (e.g., clothing items) it is often *impossible for the retailer to monitor* consumer use during the trial period, so opportunistic behavior cannot be identified at the point of return.

Our work contributes several new insights into wardrobing by considering a more realistic and nuanced representation of wardrobing behavior than in previous studies. Specifically, to our knowledge, we are the first paper to jointly examine the following elements of wardrobing: the proportion of consumers that would consider wardrobing (which we refer to as the *extent of wardrobing*), and the fraction of the product value that can be extracted by opportunistic consumers during the trial period (the *benefit of wardrobing*). As we show in our model results, both of these elements have important and sometimes surprising impacts on a retailer's optimal return policy and profits. Given their centrality to our analysis and insights, we briefly introduce each of these two elements below.

First, consider the *extent* of wardrobing in a retail context. Most analytical models of consumer returns assume that *no* consumers will consider wardrobing (see Akçay et al. (2013) and Hsiao and Chen (2012) for examples). On the other hand, in the extant wardrobing literature, it is generally assumed that *all* consumers will consider wardrobing (Chu et al. 1998, Hess et al. 1996). We suggest that the best representation of reality lies between these two extremes. Specifically, since wardrobing is not consistent with the intention of the returns policy, it may be viewed as socially unacceptable by some consumers (indeed, casual conversations about this research have elicited responses ranging from open admissions of wardrobing to expressions of outrage at the idea). Willingness to consider such actions is a function of one's psychological traits such as self-concept (Mazar and Ariely 2006) and symbolic self-completion (Rosenbaum and Kuntze 2005). Laboratory evidence has shown that individuals who emphasize the alignment between self-image and social desirability will not engage in opportunistic behavior (Mazar et al. 2008). Thus, a model of returns behavior should reflect the fact that some consumers reject wardrobing while others include it in their consideration set. Specifically, we use the term "extent of wardrobing" to denote the size of the consumer segment who will consider acting opportunistically. Note that our inquiry into the impact of the extent of wardrobing further distinguishes our work from the renting literature – since there are no social concerns with renting, renting models assume that *all* consumers consider the rental option if offered (i.e., the extent of renting is 100% of consumers).

Next, consider the *benefits* that can be realized by those who are willing to engage in wardrobing. In contrast with ordinary consumers, opportunistic consumers are willing to consume the product partially during the trial period and pay only a restocking fee (if any) in exchange for this utility. We suggest that the benefits to be gained from such behavior are largely product-specific. Consider, for example, a tuxedo vs. a casual jacket. The former is typically used on rare occasions, while the latter is intended for everyday wear. In other words, a larger fraction of the value of a tuxedo can be consumed in just a few uses, as opposed to the much slower consumption of a casual jacket. The same logic applies for many other product categories —camcorders vs. music players, sunglasses vs. eyeglasses, etc.—in which some items might be needed for use on specific, rare occasions (such as a camcorder or sunglasses on a vacation) while others provide recurring benefits over a longer period of time. This implies that the benefit of wardrobing, in addition to its extent, is an important consideration for the retailer in setting prices and refunds.

Our goal in this study is to examine how the *extent* and *benefit* of wardrobing impact a retailer's optimal pricing, refunds, and ultimately profits. This new perspective on wardrobing leads to several insights, which we briefly summarize here. First, we show that the optimal price and refund are never increasing in the extent of wardrobing. That is, as the proportion of consumers that consider wardrobing increases, the optimal response is not an increase in either price or refund. Second, in terms of profits we show that, although a larger extent of wardrobing often hurts firm profits, if the benefit of wardrobing is high, firm profit can actually *increase* when more consumers are

opportunistic. In other words, with appropriate adjustments to the price and refund, there are cases in which retailers can gain from a higher extent of wardrobing. The profit impacts of changes in the benefit of wardrobing, on the other hand, are quite different from changes in the extent. A higher benefit of wardrobing often decreases profits, but we also find cases in which profits are unaffected or even increase with the benefit of wardrobing. Since the benefit of wardrobing can be influenced by actions such as adjusting the return time window, these results have important implications for retail practice. Third, we explore the use of a price-refund menu to screen apart the opportunistic from the ordinary consumers. We find that effective screening exists only when the extent of wardrobing exceeds a threshold. That is, the retailer is no worse off to stay with the single pair of price and refund when there are not many opportunistic consumers in the market. Fourth, the profit implications of the extent and benefit of wardrobing remain qualitatively unchanged when incorporating *ex ante* heterogeneity in consumers' product valuation and when considering a retailer facing a large procurement cost of the product. Finally, we examine the scenario where the opportunistic consumers are extra careful and hence, their returns are in better condition and having higher salvage value than those made by ordinary consumers. In this case, both extent and benefit of wardrobing increase profit.

2. Literature Review

2.1. Restocking Fees, Product Returns and Wardrobing

Spurred by the real-world challenge of managing returns, the academic community has seen a growing interest in the area of consumer returns, especially from the retailer's perspective. The focus of several analytical papers, including this study, is the optimization of the restocking fee under different market and product conditions. For example, Hsiao and Chen (2012) show how the optimal restocking fee is affected by risk of product inappropriateness when some consumers value an appropriate product more than others. Shulman et al. (2009) derive the optimal restocking fee for a monopoly retailer selling multiple products, which is extended by Shulman et al. (2011) to two horizontally differentiated retailers that both allow product exchanges. Other analytical papers have focused on the joint optimization of the restocking fee and other retail decisions, such as inventory (Ketzenberg and Zuidwijk 2009) or "open-box" sales (Akçay et al. 2013). In all of these papers, it is assumed that consumer returns are driven solely by a "mismatch" or low realized product valuation for the consumer. Thus, issues related to the extent and benefit of wardrobing, in which the return policy is used opportunistically to derive short-term utility from a product, cannot be understood from these models of restocking fees and consumer returns.

In contrast, there is small but growing literature that specifically addresses the wardrobing problem by explicitly modeling such behavior and examining its impact on the retailer. This literature began with the important early works of Hess et al. (1996) and Chu et al. (1998). The models and analyses in those papers provided new insights into the impact of wardrobing on retailers, but they left open several avenues of inquiry which we address here. First, both Hess et al. (1996) and Chu et al. (1998) assume a simple dichotomous match/no match probability, identical for all consumers, as the driver of returns. In contrast, we follow recent work on consumer returns (Akçay et al. 2013, Su 2009) by incorporating a heterogeneous post-purchase valuation adjustment, with the resultant *ex post* net utility driving the return decision. The benefit of our modeling approach is that, when post-purchase valuation adjustments are modeled as continuous (as they are likely to be in reality), we can take a more nuanced view of how marginal changes in key model elements such as prices, refunds, or the benefits of wardrobing can influence wardrobing behavior and, ultimately, firm profits. Secondly, and perhaps more importantly, while the previous work on wardrobing does consider its benefits to consumers, it does not consider the *extent* of wardrobing. That is, both Hess et al. (1996) and Chu et al. (1998) assume that all consumers consider wardrobing (i.e., the extent is 100%). Since in reality there are certainly individuals who would not consider such behavior (as described in the Introduction), our model incorporates the fact that the extent of wardrobing is not necessarily 100%. By including both the benefit of wardrobing (as in previous work) and the extent of wardrobing (new in our study) as key parameters in our model, we are able to provide several counterintuitive insights into the impact of wardrobing on firm prices, refunds, and profits. In particular, our insights into the critical role that the extent of wardrobing plays in determining the impact of wardrobing on a retailer cannot be obtained from previous work in this area. For example, our model is the first to provide guidance for retailers regarding when and if efforts to combat wardrobing (i.e., influence its extent) will lead to higher profits. Finally, one recent working paper that has acknowledged the market split between opportunistic and ordinary consumers is Timoumi and Coughlan (2013), who examine retail price commitment in a multi-period setting with a fraction of wardrobing consumers. In contrast to our assumption that opportunistic

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consumers will only engage in wardrobing if it maximizes their utility, in that study, opportunistic consumers are assumed to always wardrobe. Our study complements Timoumi and Coughlan (2013) by examining a different retail problem – we do not look at price dynamics over time, but rather focus on the marginal impact of both the extent and benefit of wardrobing on retailer prices, refunds, and profits in a single-period setting.

Given the similarity of wardrobing behavior and traditional renting, in the following subsection, we provide a summary of relevant papers from the renting literature, along with a discussion of how the wardrobing context differs from this literature.

2.2. Renting Vs. Selling

Considering that wardrobing involves a short-term product usage, our Study naturally relates to the large existing literature on renting vs. selling products. However, the wardrobing context differs from a traditional renting model in several important ways. Before describing the key distinctions between our work and papers in this area, we provide an overview of several relevant papers below.

There is an active literature on renting of *information goods* (often called "pay-per-use").[2] In this literature, the primary focus is on the optimal pricing structure – pay-per-use and/or flat fee – given market conditions typically associated with information products such as software access or online movie streaming. Examples of key model features in this context include network externalities (Choudhary et al. 1998), allocation of capacity when marginal production cost is negligible (Essegaier et al. 2002), or direct usage monitoring (Sundararajan 2004) that can be used to determine usage-based fees.

There is also a large literature on renting of *durable goods* (often called "leasing"), for example, Coase (1972), Bulow (1982), Huang et al. (2001), Agrawal et al. (2012) and others – please see Waldman (2003) for a review. Key questions in this literature include how many rental usages can be expected from a durable good, and whether the profitability of a renting approach in a multi-period setting exceeds that of selling (or of operating both a sales and rental channel concurrently). An overarching theme of this literature relates to the time inconsistency problem inherent with multiple usage cycles, with a key takeaway being that leasing can be more profitable than selling since it enables the monopolist to maintain high prices over time, while with a selling model prices usually go down over time in order to generate new purchases.

Despite the similarities in terms of short-term usage behavior, there are critical contextual differences between the above literature and the wardrobing context. Most importantly, in pay-per-use and leasing papers, the only option for short-term usage is the dedicated rental channel, and there is no return option within the sales channel itself. In contrast, wardrobing occurs when the retailer accepts returns of sold items after a trial period to allow consumers to resolve their valuation uncertainties. In this retail setting, the sales price itself might also be considered by some consumers as a "rental" option (net of any refund) *within the sales channel*. Central to our analysis and results in this study is the proportion of consumers who would consider such an option. That is, a key parameter in our model is the extent of wardrobing, which only exists when there is an opportunity for consumers to return purchased items. Thus, the insights presented in this study are distinct from what might be derived from the existing work on pay-per-use or leasing.

3. Main Model

We consider a monopoly retailer selling to a market consisting of a mix of opportunistic and ordinary consumers. We first delineate how each segment makes its purchase and subsequent keep-return decisions. We then determine the resultant demand from each segment and derive total retailer profits.

3.1. Consumers

At the time of purchase, consumers face uncertainty regarding whether a product will fit their needs. This fit uncertainty (also referred to in the returns literature as valuation uncertainty) will be resolved after product trial. We assume that consumers have the same *ex ante* evaluation of the product, v_0, and their *ex post* adjustments to this initial evaluation, ε, are random draws from a common distribution (Akçay et al. 2013, Su 2009). Specifically, we let $v_0 = \frac{1}{2}$ and ε uniformly distributed between $-\frac{1}{2}$ and $\frac{1}{2}$. Putting it differently, consumers' true valuation of the product, $v = v_0 + \varepsilon$, is uniformly distributed[3] between 0 and 1. An alternative formulation with *ex ante* heterogeneity in consumer valuations is analyzed in section 5.2.

As described in section 1, consumers in our model are categorized into two types based on their willingness to opportunistically use the return policy to engage in wardrobing. Ordinary consumers use the product trial period for its intended purpose – to assess fit. They do not consider wardrobing, as they find this behavior to be inconsistent with social norms and thus unacceptable. Opportunistic consumers do not find wardrobing unacceptable, either because it is accepted by their social peers or because they have a low regard for social desirability (Mazar and Ariely 2006, Mazar et al. 2008). Thus, opportunistic consumers might not simply use the trial period to assess

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fit—instead, they consider the option of using the product intensively during the trial period. We denote the size of the segment of opportunistic consumers, that is, the extent of wardrobing, as γ. It follows that the rest of the market, $1-\gamma$, is made up of ordinary consumers.

Each consumer makes two decisions. First, she decides whether or not to buy the product. Consumers are rational and will make a purchase when the expected product valuation EV_i is greater than the price p, where i denotes consumer type. Note that, since consumers have the same *ex ante* valuation of the product, all consumers in a segment make the same purchase decision. In other words, the firm faces a step-wise demand function as in the return models of Akçay et al. (2013), Shulman et al. (2009), and Su (2009) (a complete analysis of an alternative, continuous-demand formulation is provided in section 5.2). Second, after realizing her true valuation of the product, each consumer decides whether to keep or return it. We consider the behavior of the two consumer segments in turn below.

Ordinary Consumers. Assume a price p and a refund r such that $p \geq r \geq 0$ (note that this implies that the "restocking fee" is equal to $p-r$). An ordinary consumer who has purchased will keep the product if her realized product valuation, v, is at least as high as the refund. If the opposite is true ($v < r$), she will return the product and claim the refund. Thus, her expected valuation from making the purchase is

$$EV_{or} = \mathbb{E}\max(v, r) = \int_r^1 v\mathrm{d}v + \int_0^r r\mathrm{d}v = \frac{1+r^2}{2}, \quad (1)$$

where the subscript *or* denotes ordinary consumers.

Opportunistic Consumers. As described above, some consumers consider partially consuming the product during the trial period and paying only the restocking fee $p-r$. Note that the refund amount is the same regardless of whether or not the consumer abused the return policy. This is grounded in the reality that it is very difficult for retailers to accurately assess the condition of a used product at the point of return. To our knowledge, most retail return counters make no attempt to assess condition or investigate opportunistic behavior before providing a refund. Indeed, a returns desk worker at L.L. Bean recently noted in an NPR interview that they are instructed to accept all returns, regardless of condition and without asking any questions: "at the returns desk, people are trained to keep their face and tone neutral. To not register any judgement, whatsoever" (National Public Radio 2016). Let β denote the fraction of product value consumed by these opportunistic consumers before return, $\beta \in [0, 1]$. It follows that, in exchange for the purchase price p, the wardrobing option provides a utility of $\beta v + r$ for opportunistic consumers. Thus, β captures the benefit of wardrobing. In the opportunistic segment, a consumer who has purchased the product compares the utilities of wardrobing, $\beta v + r$, and keeping, v, to decide whether to make an opportunistic return (note that opportunistic consumers *consider* wardrobing, but of course will keep the product if that is utility-maximizing). The two options have the same utility when $v = \frac{r}{1-\beta}$. As expected, when β is sufficiently large, that is, $\beta > 1-r$, opportunistic consumers will know *ex ante* that they always return, regardless of their realization of v. In this case, their expected valuation from purchasing the product is simply $EV_{op} = \mathbb{E}(\beta v + r)$, where the subscript *op* denotes opportunistic consumers. On the other hand, if β is smaller ($\beta \leq 1-r$), these consumers might, depending on their realization of v, decide to keep a purchased item, and thus we have $EV_{op} = \mathbb{E}\max(v, \beta v + r)$ instead. To summarize, we have

$$EV_{op} = \begin{cases} \mathbb{E}\max(v, \beta v + r) = \int_{\frac{r}{1-\beta}}^1 v\mathrm{d}v \\ \quad + \int_0^{\frac{r}{1-\beta}} \beta v + r\mathrm{d}v = \frac{1}{2} + \frac{r^2}{2(1-\beta)} & \text{for } r \leq 1-\beta, \\ \mathbb{E}(\beta v + r) = \frac{\beta}{2} + r & \text{for } r > 1-\beta. \end{cases} \quad (2)$$

3.2. Retailer

Given that there are two segments of consumers, with each segment having the same *ex ante* expectation of product valuation, in principle the retailer could face up to three demand scenarios – only ordinary consumers buy, only opportunistic consumers buy, or both segments buy. In other words, there might not be complete market coverage at optimality. Note that, given $0 \leq r \leq 1$, an intuitive condition that always holds at optimality (see Appendix S2), opportunistic consumers' expected product valuation is always higher than ordinary consumers' (i.e., $EV_{op} > EV_{or}$) and hence we can eliminate the case where only the ordinary segment buys. Thus, we are left with two possibilities – both segments buy ($EV_{op} > EV_{or} \geq p$) and only opportunistic consumers buy ($EV_{op} \geq p > EV_{or}$).[4]

If a product is returned by an ordinary consumer, the retailer earns restocking fee $p-r$ plus salvage value s_1 net of unit product cost c, that is, $p-r+s_1-c$. If returned by an opportunistic consumer, the retailer earns $p-r+s_2-c$ instead. Note that salvage values are independent of β in our model. We assume $s_1 > s_2$ to capture the fact that opportunistic returns are often in worse condition due to excessive wear and tear (Goering 1997, Obenberger and Brown 1976). For completeness, the case of $s_1 \leq s_2$, which might be more reflective of certain business contexts, is also analyzed in detail in section

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6.1. In considering the retailer's acquisition cost c, we assume that: 1. the product's salvage value $\{s_1, s_2\}$ does not exceed its acquisition cost c (i.e., used items are not more valuable for the retailer than new ones), and 2. the acquisition cost c is less than $\frac{1}{2}$. This second condition on c is equivalent to assuming that the retailer is able to procure the product at a cost that is lower than consumers' expected valuation of it. We remark that the assumption is made primarily for parsimony, since one limitation of our model is that the analysis becomes very cumbersome for large values of c. Taking these two assumptions together, we have $\frac{1}{2} > c > s_1$ (Su 2009, p. 598). Importantly, we remark that the upper bound on c is relaxed in section 6.2, where we show that the results remain unchanged for $c \geq \frac{1}{2}$ provided that c is not so large that participation in the market becomes unprofitable for the retailer.

Lastly, when a buyer retains her purchase, the retailer earns $p - c$. Denote the retailer's unit profit from ordinary consumers as π_{or}. Given the respective probabilities of keeping the product, $1 - r$, and returning it, r, we have $\pi_{or} = (p - c)(1 - r) + (p - r + s_1 - c)r$. Denote the unit profit from opportunistic consumers as π_{op1} for $\frac{r}{1-\beta} < 1$ and π_{op2} for $\frac{r}{1-\beta} > 1$. Recall that when $\beta \leq 1 - r$, *some* opportunistic consumers will return based on their realization of v, while when $\beta > 1 - r$, *all* opportunistic consumers know *ex ante* that they will return. Thus we have $\pi_{op1} = (1 - \frac{r}{1-\beta})(p - c) + \frac{r}{1-\beta}(p - r + s_2 - c)$ and $\pi_{op2} = p - r + s_2 - c$, with the subscripts $op1$ and $op2$ referring to the two distinct profit functions defined by the parametric conditions above.

Normalizing the market size to 1, the retailer's profit function can take one of the four forms in Table 1. The retailer chooses the price p and refund r that maximize profit subject to $p \geq r \geq 0$ (i.e., the maximum refund is 100% of price). The timing of the game is as follows:

1. The retailer chooses the price p and refund r.
2. Consumers make their purchases and realize their true valuations v.
3. Ordinary consumers keep the product if $v > r$ and return it otherwise; opportunistic consumers keep the product if $v > \beta v + r$ and return otherwise.

Table 1 Retailer Profit Function

	$r \leq 1 - \beta$	$r \geq 1 - \beta$
$EV_{op} > EV_{or} \geq p$	$\pi_1 = (1 - \gamma)\pi_{or} + \gamma\pi_{op1}$	$\pi_2 = (1 - \gamma)\pi_{or} + \gamma\pi_{op2}$
$EV_{op} \geq p > EV_{or}$	$\pi_3 = \gamma\pi_{op1}$	$\pi_4 = \gamma\pi_{op2}$

4. Results

As discussed earlier, the retailer will serve either both segments (full market coverage) or only the opportunistic segment (partial market coverage) at optimality. The results for partial market coverage are straightforward: when serving only the opportunistic consumers, the retailer tailors its price and refund for this segment and hence optimal profit increases with segment size. For technical completeness and expositional brevity, details for the partial market coverage analysis are contained in Appendix S1. In the following, we present the full market coverage results (i.e., the retailer's pricing strategy must balance the two segments), which illustrate all of our main insights. Solving the retailer's optimization problem yields the following lemma:

LEMMA 1. *The retailer's optimal price* (p^*) *and refund* (r^*) *depend on the extent* (γ) *and the benefit* (β) *of opportunism, which together define three regions of the solution space. Table* 2 *summarizes* p^*, r^*, *and optimal profit* (π^*).

Before discussing Lemma 1 in detail, we remark on the possibility of extremely high profit margins at optimality (i.e., c close to zero and p^* close to 1).[5] First, consider Cases I and II in Table 2. Optimal prices for these cases do not depend on c, but they do depend on salvage values. Recall that the salvage value of an item cannot exceed its acquisition cost (that is, $c > s_1 > s_2$). In Cases I and II, optimal prices are decreasing in both s_1 and s_2. Thus, if c is close to zero, s_1 and s_2 must also be close to zero, so optimal prices will never approach 1. Indeed, in Cases I and II, very

Table 2 Optimal Return Policies (Full Market Coverage of Main Model)

	Price p^*	Refund r^*	Profit π^*
Case I	$\frac{((\beta-1)(\gamma-1)s_1+\gamma s_2)^2}{2(\beta(2\gamma-1)+1)^2} + \frac{1}{2}$	$\frac{(1-\beta)(1-\gamma)s_1+\gamma s_2}{\beta(2\gamma-1)+1}$	$\frac{((1-\beta)(1-\gamma)s_1+\gamma s_2)^2}{2(1-\beta)(\beta(2\gamma-1)+1)} + \frac{1}{2} - c$
Case II	$\frac{((1-\gamma)s_1-\gamma^2)}{2(1-2\gamma)^2} + \frac{1}{2}$	$\frac{(1-\gamma)s_1-\gamma}{1-2\gamma}$	$\frac{(\gamma-1)^2+(\gamma-1)s_1(2\gamma+(\gamma-1)s_1)+2\gamma(1-2\gamma)s_2}{2-4\gamma} - c$
Case III	$\frac{(\beta-2)\beta+2}{2}$	$1 - \beta$	$(\beta^2 - \beta + s_2)\gamma - \frac{\beta^2}{2} + \beta + (\beta - 1)(\gamma - 1)s_1 - c$

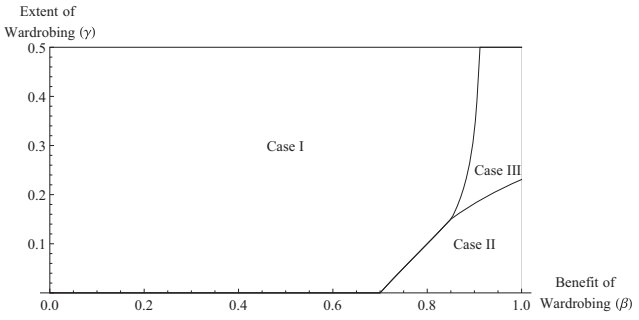
Boundary conditions for each case as well as the other two cases that define the partial market coverage are presented in Table S1 in Appendix S1.

low salvage values imply that optimal prices will be relatively low as well, as can be seen in the expressions for p^* in Table 2. Next, in Case III, the optimal price p^* depends only on β, and p^* strictly decreases with β, approaching 1 as β approaches zero. Looking at the specific conditions for Case III (see Appendix S1), we observe that Case III requires a lower bound on β, and in fact this lower bound would be quite high when s_2 is low (as must be the case for low c). Thus, in Case III, it is not possible to achieve a very high optimal price p^* when c is close to zero. In summary, extremely high profit margins (c close to zero and p^* close to 1) cannot occur at optimality.

Figure 1 provides a visual illustration of the three optimal policy regions. Note that when there are no opportunistic consumers ($\gamma = 0$), our optimal price ($p^* = \frac{1+s_1}{2}$) and refund ($r^* = s_1$) are consistent with previous literature (Akçay et al. 2013, Su 2009) given the uniform distribution of v. This finding that the profit-maximizing refund is less than the price (i.e, a partial refund is optimal) challenges the common practice of offering full refunds. Interestingly, recent developments in practice seem to suggest movement toward partial refunds, as evidenced by a recent observation by the Better Business Bureau (Owen 2014): "We have seen a trend developing where more products are hit with restocking fees, particularly those from Internet based businesses." Next, we discuss how the three distinctive types of return policies emerge from the two full market coverage demand scenarios (i.e., π_1 and π_2 in Table 1), as well as provide intuition regarding when each type makes sense to the retailer.

As explained earlier, opportunistic consumers might decide to keep a purchased item when the benefit of wardrobing, β, is low. From the retailer's perspective, low β not only makes deterring returns easier, but also profiting from them harder. As a result, the optimal return policy at low β should encourage the opportunistic consumers to keep the product. Case I describes the range of β where such a return policy is optimal. Technically, this case arises as an interior solution to π_1. As β increases, the appeal of keeping the purchased item, as opposed to wardrobing it, is reduced (recall that the probability of keep for the opportunistic segment is $1 - \frac{r}{1-\beta}$), which makes return deterrence more difficult for the retailer. In addition, since βv captures the fraction of product value that consumers can extract via wardrobing, when β is high (i.e., the wardrobing option is valued highly by opportunistic consumers) it becomes attractive for the retailer to offer a low refund and profit from wardrobing. Cases II and III together define how high the benefit of wardrobing β needs to be in order for the retailer to implement such a strategy.

Figure 1 Retailer's Optimal Return Policy Regions (Full Market Coverage of Main Model)



Note. c = 0.35, s_1 = 0.3, and s_2 = 0.15 are used.

While opportunistic consumers always wardrobe in both Case II and Case III (i.e., when the benefit of wardrobing β is very high), the difference between these two cases can be understood as follows. As evident in $\pi_{op2} = p - r + s_2 - c$, the only way for the retailer to earn more profit from wardrobing is through increasing the restocking fee $p - r$. When relatively few consumers are opportunistic (low γ, Case II), the retailer's optimal strategy is indeed to increase the restocking fee $p - r$ as the extent of wardrobing γ increases. However, if the restocking fee becomes too high, two things can occur: first, opportunistic consumers will stop wardrobing since this action is no longer utility-maximizing; second, if $r < s_2$, the retailer's unit profit of a wardrobed item ($p - r + s_2 - c$) will no longer exceed the unit profit from a kept item ($p - c$). Since the retailer wants to avoid this situation, she increases the restocking fee $p - r$ with γ up to a point, but for sufficiently high γ (Case III) the retailer stops reducing the restocking fee, and optimal price and refund are not affected by γ. Technically, while Case II is an interior solution to π_2, Case III is a corner solution obtained at $r = 1 - \beta$.

4.1. Impact of the Extent of Wardrobing

In this section, we study the profit-maximizing response to changes in the extent of wardrobing.

PROPOSITION 1.

(*a*) *Optimal refund and price are non-increasing in the extent of wardrobing.*

(*b*) *Moreover,* $\left|\frac{\partial p^*}{\partial \gamma}\right| \leq \left|\frac{\partial r^*}{\partial \gamma}\right|$.

We discuss part (a) of Proposition 1 in terms of the regions of the solution space shown in Figure 1. When the benefit of wardrobing β is low or intermediate (Case I), the retailer is unable to extract a high payment from opportunistic consumers for the wardrobing option, and it is optimal to discourage wardrobing by decreasing the refund. At the same

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time, it needs to decrease price in order to offset this less attractive refund in the minds of ordinary consumers. When the benefit of wardrobing β is very high, however, the dynamics change, as discussed in the text following Lemma 1 above. For a sufficiently high β, opportunistic consumers are willing to pay so much to engage in wardrobing that the retailer chooses to profit from wardrobing by increasing the restocking fee $p - r$. We find that this higher restocking fee is always accomplished by decreasing both price and refund, but decreasing the refund more than price (see the discussion of part (b) of the Proposition below). This occurs as long as the extent of wardrobing γ is relatively low (Case II). When both the benefit and extent of wardrobing are very high (Case III), the retailer will stop increasing the restocking fee altogether (i.e., stop adjusting the price and refund) in an effort to avoid discouraging the (highly profitable) wardrobing behavior.

The intuition for part (b) follows from the idea that the negative impact of a decrease in the refund can be offset by a price decrease in order to avoid losing ordinary consumers. At optimality, an ordinary consumer's probability of keeping the product is always larger than the probability of returning it. This is because the return probability is governed by the optimal refund amount, which is always smaller than $\frac{1}{2}$ (implying the probability of keep always larger than $\frac{1}{2}$). Thus, the negative utility impact of a refund decrease can be offset by a price decrease of a lesser magnitude since the probability of paying the price is larger than the probability of receiving the refund.

Next, we turn to the profit implications of the extent of wardrobing. For better exposition, we gather the more involved boundary definitions for the β and γ parameters in Table 3.

Table 3 Definitions for Parameter Boundaries

$\hat{\beta} = \dfrac{\sqrt{(s_1-s_2)(9s_1-s_2)}+s_2}{2s_1} - \dfrac{1}{2}$	$\dot{\beta} = \dfrac{(c-s_2)\left(\sqrt{(\alpha-1)(-\alpha+c+(\alpha-1)s_2)((\alpha-1)(c-\alpha)+(\alpha^2+1)s_2)}+\alpha(-\alpha+c+(\alpha-1)s_2+1)\right)}{c((2\alpha-1)c-2(\alpha-1)\alpha)+(\alpha-1)s_2(2(\alpha+(\alpha-1)c)-(\alpha-1)s_2)}$
$\ddot{\beta} = \dfrac{\alpha(1-c)+c-s_2}{\alpha}$	$\ddot{\beta} = \dfrac{c-s_2}{\frac{\alpha}{(1-\alpha)s_2}}$
$\hat{\gamma} = \dfrac{(\beta-1)^2-s_2(8\beta^2-12\beta+s_2+4)+(\beta-1)(2s_1(3\beta-3s_2-1)+4s_1^2)-(\beta+s_2-1)\sqrt{(\beta-1)^2+4(\beta^2-1)s_1^2-4(\beta-1)s_1(\beta+s_2+1)+s_2(6\beta+s_2-6)}}{4((\beta-1)(\beta+s_1(2\beta+s_1)-2s_2(2\beta+s_1))-s_2^2)}$	$\dot{\gamma} = \dfrac{\beta+s_1-1}{2\beta+s_1-1}$
	$\ddot{\gamma} = \dfrac{(s_1-s_2)(-1+\beta)}{s_2-s_1(1-\beta)-2s_2\beta}$

$\hat{\beta}$ in Proposition 2; $\hat{\gamma}$, $\dot{\gamma}$, and $\ddot{\gamma}$ in Proposition 3; $\dot{\beta}$ and $\ddot{\beta}$ in Proposition 8; $\ddot{\beta}$ in Proposition 9; $\dot{\gamma}$ also in Proposition 12.

Proposition 2.

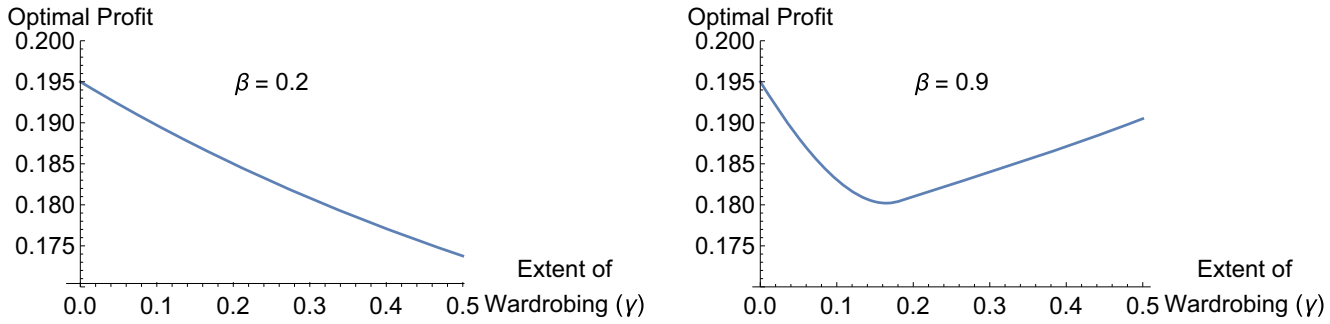
(*a*) *When the benefit of wardrobing is low* ($\beta < \hat{\beta}$), *optimal profit is decreasing in the extent of wardrobing*, γ.

(*b*) *When the benefit of wardrobing is high* ($\beta > \hat{\beta}$), *optimal profit is U-shaped in the extent of wardrobing*, γ.

Figure 2 provides a visual illustration of the above proposition. In the following, we develop the intuition behind this proposition through an approach that applies to all three optimal solution cases. Specifically, we decompose the profit impact of the extent of wardrobing γ into three parts, $\frac{\partial \pi^*}{\partial \gamma} = \frac{\partial[(1-\gamma)\pi^*_{or}+\gamma\pi^*_{op}]}{\partial \gamma} = (1-\gamma)\frac{\partial \pi^*_{or}}{\partial \gamma} + \gamma\frac{\partial \pi^*_{op}}{\partial \gamma} + (\pi^*_{op} - \pi^*_{or})$. The first two parts measure the indirect effects of γ transmitted by the segment-specific unit profits, while the last part measures the direct effect of replacing ordinary

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Figure 2 Impact of the Extent of Opportunism on Optimal Profit [Color figure can be viewed at wileyonlinelibrary.com]



Note. c = 0.35, s_1 = 0.3, and s_2 = 0.15 are used.

consumers with opportunistic ones. As the extent of wardrobing γ increases from zero, the retailer gradually shifts its focus toward profiting from the opportunistic segment by reducing the refund. As a result, the unit profit it earns from the ordinary segment decreases ($\frac{\partial \pi_{or}^*}{\partial \gamma} < 0$), while its unit profit from the opportunistic segment increases ($\frac{\partial \pi_{op}^*}{\partial \gamma} > 0$). Note that this implies that the difference in the unit profit of an opportunistic consumer and an ordinary one, $\pi_{op}^* - \pi_{or}^*$ (i.e., the *benefit from replacing an ordinary consumer with an opportunistic one*), increases as γ increases (technically, since $\frac{\partial \pi_{op}^*}{\partial \gamma} > 0$ and $\frac{\partial \pi_{or}^*}{\partial \gamma} < 0$, $\pi_{op}^* - \pi_{or}^*$ increases in γ) – a result that will be critical in the following discussion. It can also be shown that the weighted average of these two counteracting unit profit effects ($(1-\gamma)\frac{\partial \pi_{or}^*}{\partial \gamma} + \gamma \frac{\partial \pi_{op}^*}{\partial \gamma}$) is always zero. Thus, the third part of the profit impact of γ – the replacement of ordinary consumers by opportunistic ones – determines overall effect of γ on optimal profit.

Since the benefit from replacing an ordinary consumer with an opportunistic one ($\pi_{op}^* - \pi_{or}^*$) increases as γ increases, the retailer receives the largest benefit from replacing an ordinary consumer with an opportunistic one when γ is small. However, note that this "benefit" can in fact be either negative (the replacement hurts profits) or positive (the replacement helps profits). As discussed below, the sign (negative or positive) of the benefit of replacement depends critically on the benefit of wardrobing β.

When the benefit of wardrobing β is low, wardrobing is relatively unattractive, and the retailer is unable to profit from it. Thus, as the size of the opportunistic segment γ increases, the replacement of ordinary consumers with opportunistic ones hurts retailer profits. On the other hand, when β is high, opportunistic consumers are willing to pay a high price to engage in wardrobing, which implies that the retailer is able to profit more from the wardrobing through a low refund. Indeed, we see in Proposition 2 that optimal profit can increase with the opportunistic segment size (i.e., the benefit of replacing an ordinary consumer with an opportunistic one can increase to the point where this benefit is actually positive) only when β is high.[6]

To summarize, opportunistic consumers are always detrimental to the retailer's profit when wardrobing is relatively less attractive to them (low β). However, when these consumers can extract more benefit from wardrobing (high β) and hence become more profitable to the retailer, an increase in the opportunistic segment has a counter-intuitive U-shaped effect on retailer's profit. In other words, if the retailer adjusts price and refund optimally, an increase in the extent of wardrobing can increase profits in some cases.

COROLLARY 1. *When optimal profit is decreasing in the extent of wardrobing, it decreases at a decreasing rate. That is,* $\frac{\partial^2 \pi^*}{\partial \gamma^2} \geq 0$ *given* $\frac{\partial \pi^*}{\partial \gamma} < 0$.

The above finding makes an interesting practical implication for retailers. Consider the technologies available for identifying individuals who make return claims at abnormally frequent rates, such as the Verify-2 by Retail Equation Inc. (Speights and Hilinski 2005). Employing such tracking technologies, retailers can marginally reduce the number of opportunistic consumers, γ. Now, assume a retailer is in the region where its profit is negatively affected by γ. Corollary 1 provides guidance into when the marginal benefit of adopting the technology is highest. Specifically, we find that a reduction in the opportunistic segment size provides the largest payoff when size of the segment is small. Therefore, when informed by this result, retailers should adopt a more proactive approach toward combating abusive returns. In other words, the best timing for investing in the returns tracking technology is, counter-intuitively, not when the opportunistic consumer population is considerable but when it is still limited.

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4.2. Impact of the Benefit of Wardrobing

We now turn our attention to the second aspect of consumer opportunism – the benefit of wardrobing, β. Note that retailers might have some limited influence over β through adjusting the time duration of the return window – longer windows provide more opportunity to benefit from an item before returning it. Thus, β is analogous to γ in the previous section in that, while neither is a firm's decision variable, both might be influenced at the margin (β by adjusting the return time window, γ by implementing consumer tracking technologies). This limited endogeneity makes the following comparative statics particularly relevant.

PROPOSITION 3. *The marginal impact of the benefit of wardrobing (β) on profit, $\frac{\partial \pi^*}{\partial \beta}$, can be either zero or positive or negative:*

(*a*) *Optimal profit is unaffected by β for $1 - s_1 < \beta < 1$ and $0 < \gamma < \min\{\hat{\gamma}, \acute{\gamma}\}$.*
(*b*) *Optimal profit increases with β for $\frac{s_1 - s_2}{s_1} < \beta < 1$ and $\max\{\grave{\gamma}, \acute{\gamma}\} < \gamma < \frac{1}{2}$.*
(*c*) *Optimal profit decreases with β otherwise.*

In comparing Proposition 3 with Proposition 2, it is interesting to note that, while both β and γ can be considered as measures of the potential negative impact of opportunism, they affect profit in different ways. Specifically, an increase in the benefit of wardrobing, β, has a more nuanced impact – it could be either beneficial or irrelevant or detrimental to retailer's profit. Since β to a certain extent is controlled by the time length of return window, Proposition 3 offers interesting insights into when and how a longer return window exerts an indirect effect on retailer's profit.

When the extent of wardrobing γ is relatively low (part (a) of Proposition 3), the retailer tailors its return policy more towards the ordinary segment through a high refund. This high refund, if coupled with a high benefit of wardrobing β, induces the opportunistic consumers to *always engage in wardrobing*. As long as γ is low, it is too costly for the retailer to combat this wardrobing through a low refund, which would discourage the ordinary consumers (who make up most of the market) from buying (the retailer must guard against losing this segment). Thus, the retailer simply maintains a "static" return policy in this region (no changes to p or r) as β increases, and profits are unaffected. This explains why, in part (a) of the above proposition, β has no impact on optimal profit when β is high and γ is low.

Next, we consider part (b), where γ is high that is, there are many opportunistic consumers in the market. In this context, the retailer's return policy needs to be more responsive to the opportunistic segment. As β increases, wardrobing becomes more appealing, and the retailer can aggressively reduce the refund r and enjoy a higher unit profit from the opportunistic segment. The lower refund does deter ordinary consumers, but the increased profits from the opportunistic segment more than offset this loss, and the net result is that profits increase.

Lastly, in part (c) we find that increases in the benefit that consumers receive from opportunistic behavior never improve firm profits when both β and γ are small. The intuition shares some similarity with that in part (a), since again the retailer is focusing on the (larger) ordinary segment with a high refund. The difference is that, when the wardrobing option is not very appealing, opportunistic consumers do not always engage in wardrobing. As β increases, the wardrobing behavior becomes more frequent, which triggers the retailer to combat it by lowering the refund. At the same time, the retailer also needs to lower the price to keep ordinary consumers in the market. Overall, the profit loss from the ordinary segment outweighs the gain from combating wardrobing, since the level of refund is relatively generous.

5. Alternative Model Formulations

In this section, we investigate two alternative model formulations that provide additional perspectives on our problem context. For each new formulation, we discuss how our main insights change as well as new insights that emerge.[7]

5.1. Retailer Offers a Menu of Price/Refund Pairs

The existence of opportunistic consumers represents a form of information asymmetry between the retailer and the consumers. Specifically, a consumer knows her type (ordinary or opportunistic) whereas the retailer does not have this information. While a single price/refund pair offered to all consumers, as assumed in our main model, is a common approach in retail practice, in this extension we consider the possibility of the retailer offering a menu of two prices and refunds – $\{p_1, r_1\}$ and $\{p_2, r_2\}$, which could potentially be used by the retailer to separate the two consumer types and earn higher total profits. When a menu is offered, consumers will compare the expected utility obtained from each price/refund pair and choose the utility-maximizing option. For the menu to be an effective screening mechanism, each segment should prefer a different price/refund pair. The following incentive compatibility constraints specify this condition:

$$EV_{or}(r_1) - p_1 \geq EV_{or}(r_2) - p_2 \qquad (3a)$$

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$$EV_{op}(r_1) - p_1 \le EV_{op}(r_2) - p_2 \quad (3b)$$

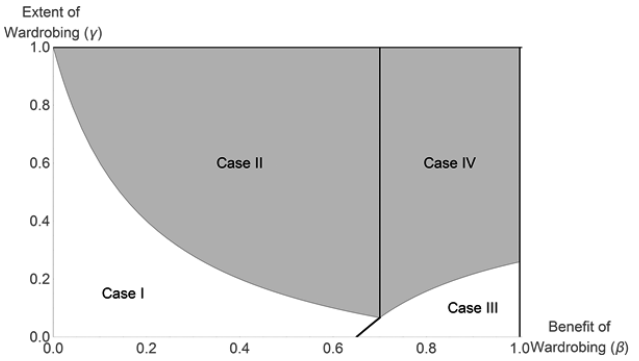
where $\{p_1, r_1\}$ is targeted to the ordinary segment and $\{p_2, r_2\}$ to the opportunistic segment. Since all consumers would prefer a low-price-high-refund choice and no consumers would prefer a high-price-low-refund choice,[8] the comparison between the two pairs should be either $p_1 \le p_2, r_1 \le r_2$ or $p_1 \ge p_2, r_1 \ge r_2$. It is straightforward to show that the latter cannot happen.[9] Thus, ordinary consumers prefer the low-price-low-refund option while opportunistic consumers prefer the high-price-high-refund option, confirming the intuition.

As discussed in section 3.2, the expected product valuation before purchase (given the same price and refund) is strictly higher for opportunistic consumers, $EV_{op} > EV_{or}$. As a result, we further know that $EV_{op}(r_1) - p_1 > EV_{or}(r_1) - p_1$. This implies, similar to the main model, the retailer either serves only the opportunistic segment or both segments. The retailer's profit function can be obtained by modifying Table 1 to reflect consumers' preference for specific price/refund pairs (See Appendix S3 for details). Solving the retailer's optimization problem yields the following:

LEMMA 2. *The retailer's optimal menu of prices and refunds depends on the extent* (γ) *and benefit* (β) *of wardrobing, which together define four regions of the solution space* (*Cases I-IV*). *Table OA.2 in Appendix S3 summarizes* $\{p_1^*, r_1^*\}$, $\{p_2^*, r_2^*\}$, π^*, *as well as the boundary conditions for each region.*

To facilitate discussion, Figure 3 depicts the four regions defined in the above lemma. We begin by remarking that, when a price/refund menu is offered, it is never optimal for the retailer to serve only the opportunistic segment. To see the intuition, consider

Figure 3 Retailer's Optimal Return Policy Regions (Alternative Formulation: Menu of Price/Refund Pairs)



Note. $c = 0.4$, $s_1 = 0.35$, and $s_2 = 0.3$ are used. The menu effectively screens the two consumer segments and hence leads to higher profits than a single price/refund pair in Cases II and IV, which are shaded.

the scenario in which the retailer sets $\{p_2, r_2\}$ to be the profit-maximizing price and refund specifically for the opportunistic segment, that is, what she would do if serving only this segment. Then, it is easy to show that the retailer can choose a pair of $\{p_1, r_1\}$ that extracts positive profit from the ordinary segment while satisfying the incentive compatibility constraint, which essentially makes serving only the opportunistic segment suboptimal.

Although the menu provides the retailer with an opportunity to screen apart the two consumer types, such screening is not always optimal. In other words, the effective screening contract does not always exist, which is formalized in the following proposition.

PROPOSITION 4. *Compared to the main model of a single price and refund, offering a menu of prices and refunds improves the retailer's profit only when the extent and benefit of wardrobing are not too low* (*Cases II and IV of Lemma* 2).

To understand the intuition behind Proposition 4, we begin by reviewing some features of our main model of a single price p^* and refund r^*. In that model, $r^*|_{\gamma \to 0} = s_1$ and r^* decreases with γ. Thus, when γ is relatively low, r^* is less than s_1 but greater than s_2, which are the profit-maximizing refunds for the ordinary and the opportunistic segments, respectively.[10]

Coming to this alternative model of a menu of price/refund pairs, note that in order to create a menu that would successfully screen the two consumer types, we need a refund targeted to the opportunistic segment, r_2, that is higher than the refund targeted to the ordinary segment, r_1. We consider the feasibility of such targeted refunds in each region of the solution space. First, consider regions in which the extent of wardrobing γ is relatively low (Cases I and III). As noted above, in the main model a low γ implies a refund r^* that is lower than the profit-maximizing refund for the ordinary segment s_1 and higher than the profit-maximizing refund for the opportunistic segment s_2. Now, to effectively screen consumers, we must replace the single r^* with two refunds, r_1 for the ordinary segment and r_2 for the opportunistic segment, with $r_2 > r_1$. However, note that any $r_2 > r^*$ results in the retailer extracting less profit from the opportunistic segment (i.e., it moves r_2 away from its profit-maximizing level of s_2), while any $r_1 < r^*$ decreases profit from the ordinary segment (i.e., it moves r_1 away from its profit-maximizing level of s_1). Therefore, in these low-γ cases the retailer cannot create a menu that effectively screens the market, and is thus no worse off to stay with the single pair of price and refund. Specifically, the optimal price and refund

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in Case I and III are similar to those in Case I and II of the main model.[11]

Second, consider the regions where the extent of wardrobing γ is relatively high (Cases II and IV). In these cases, γ has become high enough that the r^* from the main model is actually less than s_2 (recall that r^* decreases with γ). When the retailer attempts to screen the market by increasing r_2, it is moving r_2 toward its profit-maximizing level of s_2. Such an adjustment of r_2 screens the market while increasing profits, and hence the retailer is able to attain higher profits by offering a menu of price/refund pairs. Specifically, we find that the optimal refund directed to the opportunistic segment r_2^* in Case II and IV is the same as optimal refund r^* in Case IV and V of the main model (i.e., serving only the opportunistic segment).

Next, we consider whether and how each result from the main model changes under this extension. A re-examination of Proposition 1, the impact of γ on the optimal price and refund, yields the following:

Proposition 5. *When a menu of prices and refunds is more profitable than a single price and refund, the optimal price targeted to opportunistic consumers, p_2^*, is increasing in γ.*

Whenever the retailer can increase profits with a menu of prices and refunds (Cases II and IV of Lemma 2), it charges a refund invariant to the extent of wardrobing (see Table OA.2). Intuitively, as the size of the opportunistic segment γ increases, the retailer's pricing structure should generate a higher unit profit from these consumers, as in the main model. Since the refund does not change, it is straightforward to see that the retailer increases price, p_2^*, and extracts more unit profit from the opportunistic segment.

In terms of Proposition 2, the impact of γ on profits, we now find the following:

Proposition 6. *When a menu of prices and refunds is more profitable than a single price and refund, optimal profit is U-shaped in the extent of wardrobing γ for all values of β.*

Unlike in the main model, where for small β we observe profits strictly decreasing in the extent of wardrobing γ, we find in this alternative model that, regardless of the value of β, optimal profit is *always* U-shaped in γ. Two reasons together contribute to the outcome of this comparison. First, when the retailer is able to screen the opportunistic consumers from the ordinary ones, it can tailor the price and refund to each segment to a larger extent than in the main model. This consequently improves the retailer's ability to profit from the presence of opportunistic consumers. Second, the effect of such improved targeting on profit is most salient when the extent of wardrobing is large. This is because as γ increases, the retailer both earns a higher unit profit from the opportunistic consumers and faces a larger proportion of them. Recall from the discussion of Proposition 2 that when the benefit of wardrobing is small and hence opportunistic consumers are relatively less profitable, replacing an ordinary consumer with an opportunistic one always results in a loss to the retailer. Thus, profit is strictly decreasing in γ for small β. With the menu of prices and refunds, the improved targeting to and the resultant extra profit from the opportunistic consumers offset the decrease in the profit from the ordinary consumers when the extent of wardrobing is high enough. As a result, U-shaped relationship between γ and π^* also happens at small β.

Lastly, we note that the results in Corollary 1 and Proposition 3 from the main model are unaffected by this model extension (analysis available from the authors):

Remark 1. Corollary 1 and Proposition 3 remains unchanged under this model formulation.

5.2. Continuous Demand Formulation

In this extension, we present an alternate formulation of our model which results in a continuous demand function. This enables us to examine within-segment marginal demand changes that do not appear in discrete demand models such as the one used in our main analysis.

Recall that a consumer's valuation of an experience good, v, is typically separated into the core part (observed *ex ante*) and the uncertain part (resolved *ex post*). Since consumer returns are driven by the *ex post* uncertainty, studies of consumer returns have often assumed a simple homogeneous core valuation v_0 and a continuously distributed uncertainty ε (e.g., Hsiao and Chen 2012, Shulman et al. 2009). In our main model, we also follow this approach.

However, the homogeneous v_0 assumption also presents a potential drawback because marginal changes in price or refund will not change demand until a threshold is reached, at which point an entire segment of consumers will switch their initial purchase decision. To provide another perspective, in this extension we consider a continuously distributed core valuation. This within-segment heterogeneity in core valuations leads to demand that is a continuous function of price and refund. Regarding *ex post* uncertainty, recall that in our main model this was also continuously distributed. Given the richer modeling of core valuations in this section, a simpler

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representation of *ex post* uncertainty is needed to maintain tractability. Thus, we make the simplifying assumption here that the product is a match to the consumer's needs with probability α and it is a mismatch with probability $(1-\alpha)$.[12] This match/mismatch (binomial) representation of uncertainty in experience goods is also a common abstraction of the reality in retail (e.g., Hsiao and Chen 2012, Shulman et al. 2009). If the product matches the consumer's needs (α probability), she will obtain a value of v_0, where v_0 is uniformly distributed between 0 and 1. Note that this is the same definition of v_0 as in the main model (i.e., ex ante core product valuation). While v_0 was set to $\frac{1}{2}$ in the main model, it varies from consumer to consumer between 0 and 1 here. If there is a mismatch ($1-\alpha$ probability), the consumer obtains zero value. Note that under this formulation, all consumers will return the product in the case of a mismatch. Furthermore, if an ordinary consumer chooses to make a purchase ($\alpha v_0 > p$), she will always keep the product if it is a match.

An ordinary consumer's expected valuation from making the purchase is $EV_{or} = \alpha v_0 + (1-\alpha)r$. Solving $EV_{or} = p$, we obtain the indifferent ordinary consumer at $\dot{v}_0 = \frac{p-(1-\alpha)r}{\alpha}$. For an opportunistic consumer who bought the product and finds a match, she compares "wardrobe" option ($\beta v_0 + r$) with the "keep" option (v_0) to decide which action to take. Denote the opportunistic consumer indifferent between wardrobing and keeping as $\bar{v}_0 = \frac{r}{1-\beta}$. Specifically, those with $v_0 < \bar{v}_0$ will engage in wardrobing (always return) and those with $v_0 > \bar{v}_0$ will only return if there is a mismatch. In effect, the latter group of opportunistic consumers will behave in the same way as ordinary consumers, and thus have the same *ex ante* expected valuation as ordinary consumers, while the former group of opportunistic consumers will have an expected valuation of $EV_{op} = r + \alpha\beta v_0$, which equals p at $\ddot{v}_0 = \frac{p-r}{\alpha\beta}$.

Demand from ordinary consumers is straightforward – those with $v_0 > \dot{v}_0$ will buy, and they will return the product only if there is a mismatch. In contrast, the opportunistic consumers who have bought the product might act like ordinary consumers (i.e., return only if there is a mismatch) or they might engage in wardrobing. In Appendix S4, we show that the retailer faces four potential demand scenarios characterized by $\bar{v}_0 < \dot{v}_0 < \ddot{v}_0$, $\ddot{v}_0 < \dot{v}_0 < \bar{v}_0 < 1$, $\ddot{v}_0 < \dot{v}_0 < 1 < \bar{v}_0$, and $\ddot{v}_0 < 1 < \dot{v}_0 < \bar{v}_0$ (see Figure 4 for illustration; corresponding profit functions in Appendix S4). From that figure one can see that the total demand is governed by the positions of $\dot{v}$ and $\ddot{v}$ on the 0–1 continuum. Since both $\dot{v}$ and $\ddot{v}$ are increasing in price and decreasing in refund, total demand decreases with price and increases with refund. Consistent with the main model, the retailer earns $p-c$ for each kept product, $p-r+s_1-c$ for each return made by an ordinary consumer, and $p-r+s_2-c$ for each return made by an opportunistic consumer.

We make two simplifying assumptions to maintain tractability. First, we assume salvage value of an ordinary return equals unit cost, $s_1 = c$, and hence $0 < s_2 < c < \frac{1}{2}$. Second, α is assumed to be high enough ($\alpha > \frac{c-s_2}{1-s_2}$) to ensure the retailer earns a positive profit within the whole spectrum of γ and β. Solving the retailer's optimization problem yields the following:

LEMMA 3. *The retailer's optimal price* (p^*) *and refund* (r^*) *depend on the extent* (γ) *and the benefit* (β) *of wardrobing, which together define four regions of the solution space. Table OA.3 in Appendix S4 summarizes* p^*, r^*, *and optimal profit* (π^*), *as well as the boundary conditions for each region.*

Next, we consider how each result from the main model changes under this extension, starting with a re-examination of Proposition 1, the impact of γ on the optimal price and refund:

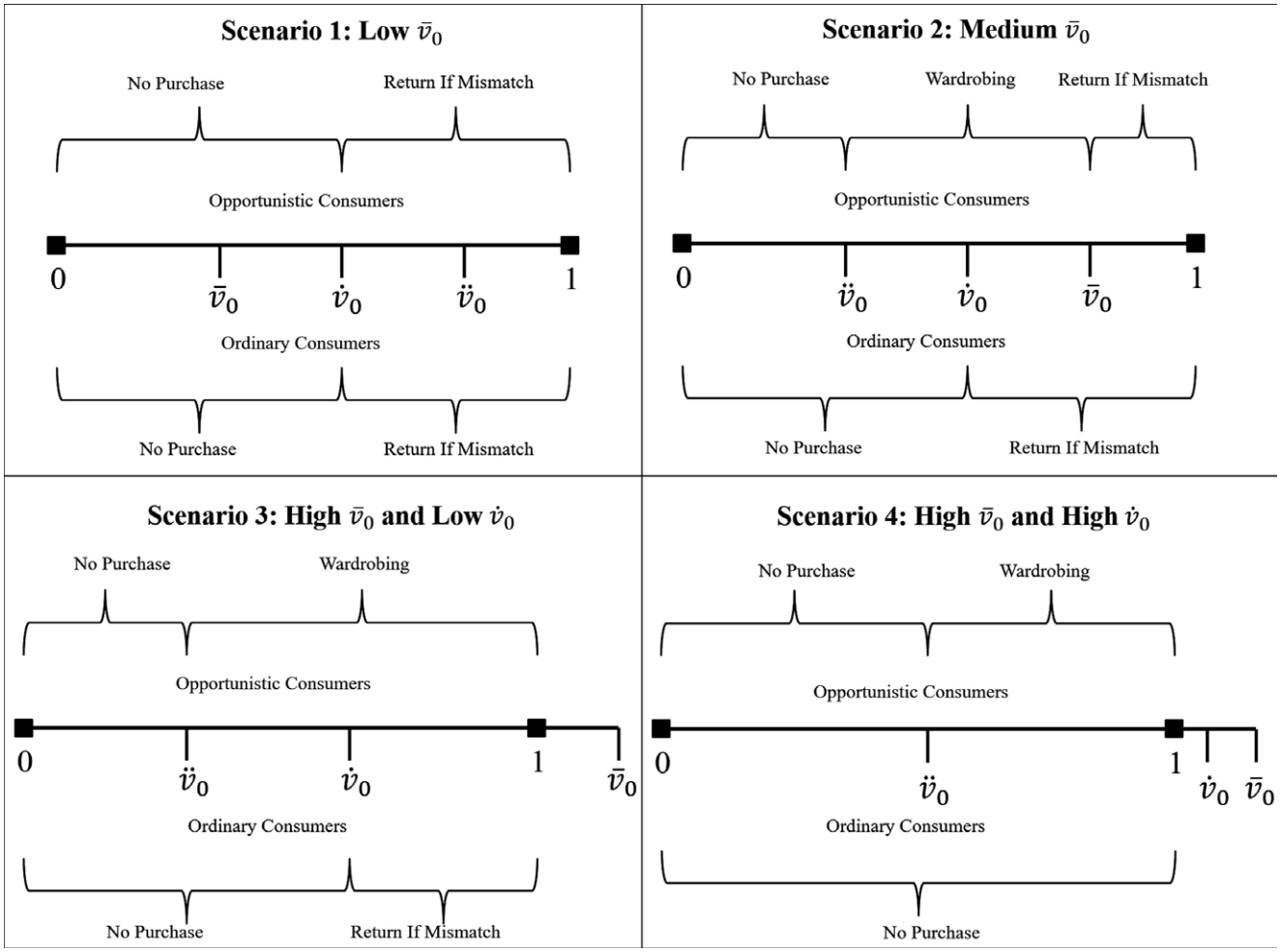
PROPOSITION 7. *When the demand function is continuous, optimal price and refund is non-decreasing in the extent of wardrobing* γ. *Moreover,* $\frac{\partial p^*}{\partial \gamma} \geq \frac{\partial r^*}{\partial \gamma}$.

The above result is partially in contradiction to the price and refund impacts of γ in our main model. This is driven by the fact that price and refund only affect consumers' *ex ante* behavior in this model. Specifically, since uncertainty is captured with the probability of match, which is unaffected by price and refund, a consumer does not change her keep-or-return behavior based on p and r in this extension – she simply makes her decision based on match/mismatch. In other words, under this formulation price and refund are only considered in the initial purchase decision. This is in contrast to our main model, where p and r are drivers of the *ex post* return decision. Thus, in the following, we build our intuition by focusing on consumers' *ex ante* behavior, discussing each of the four cases in Table OA.4 in turn.

When β is low (Case I in Table OA.4), there is not much to gain from wardrobing, and opportunistic consumers will only return in the case of a mismatch. Thus, an increase in the opportunistic segment size simply reduces the average salvage value from the retailer's perspective, and we know from the classic monopoly model that optimal price increases with unit cost (Tirole 1988). When the benefit of wardrobing β is intermediate (Case II), some opportunistic buyers engage in wardrobing ($v_0 < \bar{v}$), while others will only return if there is a mismatch ($v_0 > \bar{v}$). Since price does not affect demand from the latter opportunistic consumers (i.e., $\bar{v}$ is independent of p), the

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Figure 4 Demand Scenarios with Continuously Distributed Core Valuation



retailer exploits this fraction of the opportunistic segment by increasing price as γ increases. To offset the negative effects of this price increase to the ordinary segment, it also increases refund. Finally, when the benefit of wardrobing is large (Cases III and IV), opportunistic consumers value the wardrobing option very highly, and it is optimal for the retailer to allow all opportunistic consumers to wardrobe and thus profit from the restocking fee $p - r$. In Case IV, there exists a pair of price and refund that maximizes profit from both segments, and hence the optimal return policy does not change with γ. Such a price-refund pair does not exist in Case III. Therefore, the retailer sets a refund $r^* = 1 - \beta$, which is just enough to make the opportunistic buyers always engage in wardrobing, while increasing price to earn more profit from this segment as it grows.

In terms of the impact of γ on profits, we now find the following:

PROPOSITION 8. *When the demand function is continuous, the impact of the extent of wardrobing γ on profit can be negative, U-shaped, or positive:*

(*a*) *Optimal profit is decreasing in the extent of wardrobing given* $0 < \beta < \dot{\beta}$.
(*b*) *Optimal profit is U-shaped in the extent of wardrobing given* $\dot{\beta} \le \beta < \ddot{\beta}$ *if* $\alpha \ge \frac{c-s_2}{c}$ *and* $\dot{\beta} \le \beta < 1$ *if* $\alpha < \frac{c-s_2}{c}$.
(*c*) *Optimal profit is increasing in the extent of wardrobing given* $\ddot{\beta} \le \beta < 1$ *if* $\alpha \ge \frac{c-s_2}{c}$.

Comparing the above results and Proposition 2, we observe the only difference is the addition of one more possibility – optimal profit increases with γ, which happens when the benefit of wardrobing is very large. To explain this, the approach that we have used to build intuition for the main model continues to be valid here. That is, $\frac{\partial \pi^*}{\partial \gamma}$ can be decomposed into three parts, two of them sum to zero (i.e., $(1-\gamma)\frac{\partial \pi^*_{or}}{\partial \gamma} + \gamma \frac{\partial \pi^*_{op}}{\partial \gamma} = 0$), and hence direction of $\frac{\partial \pi^*}{\partial \gamma}$ is determined by $\pi^*_{op} - \pi^*_{or}$. It happens both in the main model and here that if β is large enough, it becomes optimal for the retailer to allow opportunistic consumers always return. Yet, when the retailer is able to adjust demand

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within segments, it extracts more profit from the opportunistic consumers who engage in wardrobing. In fact, it is possible that the optimal unit profit from this segment is always larger than that from the ordinary segment, $\pi^*_{op} > \pi^*_{or}$. As a result, we observe the addition of Proposition 8*c*.

Additionally, $\frac{\partial^2 \pi^*}{\partial \gamma^2} > 0$ always holds in Cases I-III and hence Corollary 1 continues to hold.

REMARK 2. Corollary 1 remains unchanged under this model formulation.

Finally, in terms of the impact of β on profits, we now find the following:

PROPOSITION 9. *When the demand function is continuous, the marginal impact of the benefit of wardrobing β on profit, $\frac{\partial \pi^*}{\partial \beta}$, can be either zero or positive:*

(*a*) *Optimal profit is unaffected by β for $0 < \beta < \ddot{\beta}$.*
(*b*) *Optimal profit increases with β for $\ddot{\beta} \leq \beta < 1$.*

The difference between the above results and Proposition 3 is the disappearance of negative marginal impact, $\frac{\partial \pi^*}{\partial \beta} < 0$. This change is related to the binomial stylization of product uncertainty. When uncertainty is continuous as in the main model, an increase in β will lead to a higher probability of abusing the return policy for all opportunistic consumers. However, under this formulation a higher β only shifts some opportunistic consumers' preference to wardrobe (those close to $\bar{v}$), and thus β should be less harmful to the retailer. From Proposition 3, we see that $\frac{\partial \pi^*}{\partial \beta} < 0$ happens at small β. In this case, if β is small, the retailer is able to set a price and refund such that no opportunistic consumers will wardrobe, and therefore the negative impact of β is eliminated.

6. Extensions

In this section, we investigate two extensions to our main model – careful opportunistic consumers and high unit cost. For each one, we discuss how our main insights are impacted by the extension as well as new insights that emerge.

6.1. Careful Opportunistic Consumers ($s_1 \leq s_2$)

In the main model, we assume that, due to excessive wear and tear, the salvage value of an ordinary return is greater than an opportunistic return, that is, $s_1 > s_2$. As an alternative, it is also possible that, since opportunistic consumers know that they will return the product, they might be more careful with it in order to ensure that the retailer will accept the return.[13] In our model, this would imply that $s_1 \leq s_2$. Thus, the appropriate assumption on the relationship between s_1 and s_2 will depend on factors such as the stringency of the product inspection process upon return. For completeness, in this extension, we examine how the retailer's pricing strategy and profit will be affected when $s_1 \leq s_2$ (proofs in Appendix S5).

In the following, we consider how each result from the main model changes under this extension, starting with impact of the extent of wardrobing on optimal price and refund:

PROPOSITION 10. *When $s_1 \leq s_2$ and both segments are served, optimal price and refund are increasing in the extent of wardrobing for $0 < \beta \leq \min\{\frac{s_2 - s_1}{s_1}, 1 - s_1\}$ and non-increasing otherwise. Moreover, $p^* - r^*$ is non-decreasing in γ.*

We exhibit the above results in a way such that comparison with Proposition 1 is easier. More precisely, the change happens at small β (within Case I), where the impacts of γ on p^* and r^* are now positive. To explain the intuition behind these results, we revisit the characteristics of the opportunistic consumers. Under our modeling framework, the opportunistic segment will excessively consume the product if they decide to abuse the return policy, which is captured by the benefit of wardrobing, β. This leads to two consequences: (1) the opportunistic consumers return more often, and (2) they have a higher expected product valuation. When coupled with a lower salvage value ($s_2 < s_1$), the former imposes a negative profit impact to the retailer. The latter, on the other hand, allows the retailer to extract extra surplus by effectively coordinating its refund and price. When $s_2 \geq s_1$, the negative aspect of the opportunistic consumers is essentially eliminated. In other words, the retailer's focus now becomes how to extract more profit from the opportunistic behavior rather than deterring it. As the opportunistic segment size grows, the retailer could profit from these consumers by either encouraging more returns (increase r) or earning more unit profit from each return (decrease r). When β is small, opportunistic consumers' preference to return is relatively low. Thus, the retailer increases r to induce more returns. When β is large, their preference to return is already strong. The retailer decreases r to gain more unit profit from each returned item. Different from changes in the price and refund impacts, we still observe the restocking fee, $p^* - r^*$, to be non-decreasing in γ.

Next we turn to the profit impacts of the two wardrobing constructs:

PROPOSITION 11. *When $s_1 \leq s_2$ and both segments are served, optimal profit is always increasing in the extent of wardrobing γ.*

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PROPOSITION 12. *When $s_1 \leq s_2$ and both segments are served, marginal impact of the benefit of wardrobing on profit, $\frac{\partial \pi^*}{\partial \beta}$, can be either zero or positive:*

(*a*) *Optimal profit is unaffected by β for $1 - s_1 < \beta < 1$ and $0 < \gamma < \acute{\gamma}$.*
(*b*) *Optimal profit increases with β otherwise.*

We present the above two new results together since they convey a common message – the negative profit impacts of the extent and benefit of wardrobing disappear if salvage value from an opportunistic return is higher than that from an ordinary return. Given our discussion on the characteristics of the opportunistic consumers above, it is straightforward to see that if the opportunistic product consumption adds value to consumer utility but does not harm the condition of the returns, it is in the interest of the retailer to serve more such consumers. A more precise yet technical explanation is given at the end of Appendix S5. As a result, more opportunistic consumers increase the retailer's profit. The reason that a larger benefit of wardrobing does not harm the retailer follows a similar logic: while a higher β leads to more returns, the retailer does not necessarily absorb any negative impact since salvage value of the opportunistic returns are higher. Last, we note that $\frac{\partial^2 \pi^*}{\partial \gamma^2} > 0$ remains valid in the regions where profit is decreasing in γ, which gives the following remark:

REMARK 3. Corollary 1 remains unchanged under this model formulation.

6.2. Large Unit Cost

In this extension, we consider what happens to the retailer's optimal strategy for large unit costs, that is, $c \geq \frac{1}{2}$. We begin by noting the invariance of our result for larger c up to a threshold:

PROPOSITION 13. *All results from the main model (Propositions 1–3 and Corollary 1) are unchanged for larger c, provided that it remains profitable for the retailer to participate in the market, which is always the case when $c \leq \frac{1+s_2^2}{2}$.*

When $c < \frac{1}{2}$, it can be easily verified that π^* is always positive. When c is higher, there is a threshold above which the retailer will start to face negative optimal profits and hence would rather not sell the product at all. Given our central interest in β and γ (i.e., the two opportunism constructs), we find a threshold of c, denoted by $\bar{c}$, such that $\pi^* > 0$ holds across the entire parameter space of $0 < \beta < 1$ and $0 < \gamma < 1$ as long as $c < \bar{c}$. In other words, $\bar{c}$ is the level of unit cost above which the retailer will start to face negative profits in some cases of β and γ. In Appendix S6, we prove that $\bar{c} = \frac{1+s_2^2}{2}$, which is higher than the $\frac{1}{2}$ condition used in the main model. We also show that our main conclusions do not change when $\frac{1}{2} < c < \bar{c}$, so our results extend directly up to the $\bar{c}$ threshold.

For even higher unit costs ($c > \bar{c}$), it can be optimal for the retailer not to sell any products for some values of γ and β. Exact conditions under which this nonparticipation becomes optimal for $c > \bar{c}$ are too complex for meaningful interpretation. Instead, an extensive numerical analysis of this context is reported in Appendix S6. As expected, we observe non-participation occupies a larger area in the solution space as c further increases. In addition, the numerical results are in general consistent with our propositions in the main model.

7. Conclusion

Our research contributes to the retail operations literature by incorporating the behavior of opportunistic consumers into a retailer's optimal return policy decisions. By considering two key characteristics of wardrobing – its extent and its benefit – in our analytical framework, we provide several counter-intuitive and novel managerial insights. We first demonstrate that the monopolistic retailer's optimal response to an increase in the extent of wardrobing is not always to decrease the refund. When the extent of wardrobing is low, but the benefit of wardrobing is high, it is in fact better to maintain a "static" return policy. We also show that when a decrease in refund is necessary, it should always be accompanied by a price decrease of a smaller magnitude.

Our analysis of the profit impact of the extent of wardrobing confirms the common expectation that an increase in consumer opportunism will hurt firm profits, but only when the benefit of wardrobing is low. By examining the second order effects, we refine this conventional wisdom by showing that an increase in the extent of wardrobing has the most detrimental impact on profits when the segment of opportunistic consumers is currently small. More interestingly, we also show that, if the retail context involves both a high benefit and a high extent of wardrobing, the retailer's optimal profit can actually *increase* with the size of the opportunistic segment. This implies that actively working to reduce opportunism might not increase profits in certain business contexts. Lastly, our analysis of impact of changes in the benefit of wardrobing reveals that its profit impact can be zero, positive, or negative, and we define the conditions under which each of these is the case. These results suggest that there are cases in which the firm might actually prefer to increase the benefit of wardrobing (e.g., by adjusting the length of the trial period).

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Following our main analysis, all model insights were reassessed using two alternative formulations of our problem context. First, we developed a model in which the retailer might address the adverse selection issue arising from wardrobing by offering a menu of price/refund pairs. We find that the menu improves the retailer's profit only when the extent of wardrobing exceeds a threshold. In other words, effective screening does not exist when there are not many opportunistic consumers and the retailer is no worse off staying with the single pair of price and refund. Secondly, we considered a setup with *ex ante* consumer heterogeneity, resulting in a more nuanced model of initial consumer demand. We demonstrate the robustness of our main insights and comment on the intuition behind the changes in results. In an additional extension, we show that the retailer's profit is always increasing in the extent and benefit of wardrobing if opportunistic consumers are extra careful and their returns have higher salvage value. Last, our results are generalized to situations where the retailer faces high procurement costs.

While our model captures the essential elements of the consumer returns context with wardrobing, there are other aspects that could be considered in future research. In an effort to discourage policy abusers, some retailers limit the maximum dollar value returnable by a single consumer within a period of time. For example, Best Buy explicitly states on their website that they reserve the right to refuse further returns if the consumer's return volume is considered "excessive." Evaluating the effectiveness of this practice requires a multi-period framework, which is an interesting area of future inquiry. Other strategies to discourage wardrobing, such as in-store sampling or even (where feasible) usage monitoring during the trial period, are also left to future research. Finally, while a direct extension of our model to a competitive context is not tractable, a model of competition using a simplified analytical framework is a promising research direction.

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Notes

[1]In this study, we will refer to the behavior as *wardrobing* and the consumers who are willing to consider such behavior as *opportunistic*.

[2]We thank the review team for suggesting the parallels between our work and this literature.

[3]The assumption that a consumer's post-trial product evaluation follows a uniform distribution is consistent with previous analytical models of consumer returns (see Akçay et al. (2013) and Shulman et al. (2011) for examples). As a robustness check, we have also used a truncated normal distribution to numerically examine the validity of our results. There is no qualitative change under such alternative distributional assumption. This analysis is available from the authors upon request.

[4]Note that, when only opportunistic consumers buy, the retailer pursues a strategy of selling only short-term use. In other words, we find that a focus on short-term users can be optimal, even when a separate rental channel does not exist.

[5]We thank an anonymous reviewer for suggesting this discussion.

[6]Although our analysis is focused on the case where, in general, the salvage value after opportunistic use is lower than after regular use ($s_1 > s_2$), we remark that, if $s_1 \leq s_2$, firm profits always increase in the extent of wardrobing. This is because, with equal or less salvage values, the downside of opportunistic behavior from the firm's perspective is eliminated. Formal discussion for this case is in section 1.

[7]We thank the review team for suggesting these alternative approaches to the problem and the additional insights they facilitate.

[8]Technically, $p_1 < p_2$ and $r_1 > r_2$ lead to $EV(r_1) - p_1 \geq EV(r_2) - p_2$ for both consumer segments and hence violating the incentive compatibility constraints.

[9]If $p_1 \geq p_2$ and $r_1 \geq r_2$ were true then $EV_{or}(r_1) - p_1 < EV_{or}(r_2) - p_2$ and $EV_{op}(r_1) - p_1 > EV_{op}(r_2) - p_2$. Arranging terms, we obtain $EV_{or}(r_2) - EV_{or}(r_1) > p_2 - p_1 > EV_{op}(r_2) - EV_{op}(r_1)$, which leads to $EV_{op}(r_1) - EV_{or}(r_1) > EV_{op}(r_2) - EV_{or}(r_2)$. However, examining the marginal change in $EV_{op} - EV_{or}$ with respect to refund, we know that $\frac{\partial EV_{op} - EV_{or}}{\partial r} > 0$, which contradicts with $EV_{op}(r_1) - EV_{or}(r_1) > EV_{op}(r_2) - EV_{or}(r_2)$ under $r_1 < r_2$.

[10]s_2 applies when $\beta < 1 - s_2$. When $\beta > 1 - s_2$, replace s_2 with $1 - \beta$ and the intuition is otherwise the same.

[11]More precisely, they are identical except that $\{p_2^*, r_2^*\}$ in Case III is on a continuum.

[12]While the case where *both* core valuation *and* uncertainty are continuously distributed is not tractable analytically, we provide extensive numerical examination in Appendix S4.

[13]We thank an anonymous reviewer for suggesting this alternative.

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Supporting Information

Additional supporting information may be found online in the supporting information tab for this article:

Appendix S1: Partial Market Coverage Results.

Appendix S2: Proofs for Main Model.

Appendix S3: Proofs for Alternative Formulation: Menu of Price/Refund Pairs.

Appendix S4: Proofs for Alternative Formulation: Continuous Demand Function.

Appendix S5: Proofs for Extension: Careful Opportunistic Consumers ($s_1 \leq s_2$).

Appendix S6: Proofs and Additional Results for Extension: Large Unit Cost.

Appendix S7: Numerical Results: Continuous Core and Uncertainty.

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